Sports Medicine
https://doi.org/10.1007/s40279-020-01400-x

REVIEW ARTICLE

Recommendations for Hamstring Function Recovery After ACL Reconstruction

Matthew Buckthorpe[1,2,3] · Furio Danelon[4] · Giovanni La Rosa[2] · Gianni Nanni[5] · Matthew Stride[3] · Francesco Della Villa[2]

Accepted: 28 November 2020
© The Author(s), under exclusive licence to Springer Nature Switzerland AG part of Springer Nature 2020

Abstract

It is important to optimise the functional recovery process to enhance patient outcomes after major injury such as anterior cruciate ligament reconstruction (ACLR). This requires in part more high-quality original research, but also an approach to translate existing research into practice to overcome the research to implementation barriers. This includes research on ACLR athletes, but also research on other pathologies, which with some modification can be valuable to the ACLR patient. One important consideration after ACLR is the recovery of hamstring muscle function, particularly when using ipsilateral hamstring autograft. Deficits in knee flexor function after ACLR are associated with increased risk of knee osteoarthritis, altered gait and sport-type movement quality, and elevated risk of re-injury upon return to sport. After ACLR and the early post-operative period, there are often considerable deficits in hamstring function which need to be overcome as part of the functional recovery process. To achieve this requires consideration of many factors including the types of strength to recover (e.g., maximal and explosive, multiplanar not just uniplanar), specific programming principles (e.g., periodised resistance programme) and exercise selection. There is a need to know how to train the hamstrings, but also apply this to the ACLR athlete. In this paper, the authors discuss the deficits in hamstring function after ACLR, the considerations on how to restore these deficits and align this information to the ACLR functional recovery process, providing recommendation on how to recover hamstring function after ACLR.

✉ Matthew Buckthorpe
matthew.buckthorpe@stmarys.ac.uk

1 Faculty of Sport, Health and Applied Science, St Mary's University, Twickenham, London TW1 4SX, UK

2 Isokinetic Medical Group, FIFA Medical Centre of Excellence, Education and Research Department, Bologna, Italy

3 Isokinetic Medical Group, FIFA Medical Centre of Excellence, London, UK

4 Isokinetic Medical Group, FIFA Medical Centre of Excellence, Milan, Italy

5 Isokinetic Medical Group, FIFA Medical Centre of Excellence, Bologna, Italy

Published online: 17 December 2020

Key Points

A knee flexor maximal strength deficit after ACLR is a strong risk factor for ACL re-injury.

Overcoming deficits in hamstring function after ACLR is essential for optimal outcomes, satisfactory return to sport and re-injury risk reduction.

Most of the information concerning the hamstrings is devoted to hamstring injury prevention and hamstring muscle injury rehabilitation, with a lack of information on hamstring rehabilitation after ACLR.

Most training recommendations are focused on uninjured athletes and so needs to be adapted for the injured athlete.

Understanding hamstring training considerations and applying this to the ACLR patient as part of the ACL functional recovery process is essential.

1 Introduction

The outcomes after anterior cruciate ligament reconstruction (ACLR) need improving, with fewer people than ideal returning to sport [1] and even less returning to performance [2]. A particular concern after ACLR, is the high rate of ACL and knee re-injuries after return-to-sport (RTS), particularly amongst young athletes (~30%) [3, 4]. RTS after injury is a complex, multifactorial process and requires a biopsychosocial approach [5]. Current opinion is that optimise patient outcomes after ACLR, there is a need to optimise the rehabilitation approach. There is however, no consensus on rehabilitation after ACLR, despite considerable effort in recent years to clarify and optimise the process [6–11]. One issue in clinical practice is the large disconnect between research and practice, thought due to ineffective implementation of evidence-based findings [12]. It is also well recognised that if we are to truly impact individual patients, a stronger focus on research implementation is needed to translate efficacious rehabilitative and preventive methods into practice [13–16].

One important piece of the complex puzzle of ACL rehabilitation, is restoring knee flexor muscle function. The hamstring muscles are vitally important for the knee. During forceful dynamic movements, coactivation of the hamstrings is important to provide dynamic knee joint stabilisation and to prevent excessive ACL shear forces [17, 18]. Thus, the hamstring muscles are considered ACL-agonists. ACL injury, the resultant surgery and reduced functionality after surgery significantly impact on hamstring function, with deficits of nearly 50% reported at 4 weeks after ACLR with hamstring tendon autograft (HG) [19]. Restoring hamstring function is a key aspect of the functional recovery process after ACLR [8, 9]. Unfortunately, deficits in knee flexor strength can be high at the time of RTS (0–20%) [20–24] and even for many years after ACLR [25, 26]. Although deficits in knee flexor strength are typically less than that for the knee extensors [27, 28], even small deficits in knee flexor strength can be detrimental to injury risk upon RTS. In particular, within a group of professional football players, Kyritsis et al. [29] reported a 10.6-fold increased risk of ACL re-injury upon RTS, for each additional 10% deficit in knee flexor to extensor strength ratio. Furthermore, a history of severe knee injury (including ACL injury) increases the risk of a future hamstring strain injury (HSI) [30]. Although there are multiple risk factors for HSIs, likely all inter-relating in a complex manner [31, 32], including previous hamstring history [33–35], age [33–39], hamstring muscle architecture [40–43], lumbo-pelvic hip stability [34, 44–48] and training load [49–54], amongst others, the increased risk of HSI after severe knee injury is likely in part due to the altered hamstring strength. After HSI, those athletes re-injured upon RTS were ~ 14% weaker compared to those that remained injury free when assessed prospectively [55]. So, assessing and treating knee flexor strength deficits is a major element of the ACLR functional recovery process [8, 9].

The aim of this paper is to translate the research on hamstring function, largely on hamstring strength training for performance, rehabilitation and injury prevention and then demonstrate how this can be applied to the ACLR patient. In doing so, identifying the gaps in literature we currently clearly have with hamstrings rehabilitation in ACLR populations. We discuss important alterations of the hamstring muscles after ACLR, their implications and considerations in programme design to overcome these deficits. Finally, we demonstrate how to apply this information to the ACLR patient, considering the specifics of the pathology and associated surgery. It is hoped this information supports translation of research into practice and supports more optimised hamstring recovery after ACLR, aiding in improving global patient outcomes.

2 Hamstring Function Changes After ACL Injury and ACLR: What, Why and Their Implications

2.1 Hamstring Deficits After ACLR and Their Implications

Deficits in hamstring function are common after ACLR, irrespective of the surgical procedure [28, 29, 56]. Typical deficits in knee flexor maximal strength are between 0 and 20% across all contraction modes, at the time of RTS and in the years after RTS [20–24]. In a sample of more than 4000 patients, it was shown that the proportion of patients achieving a knee flexor limb symmetry index (LSI) ≥ 90% was 47% [28]. More pronounced deficits in knee flexor strength are apparent after ACLR with HG [57, 58]. Cristiani et al. [28] reported that patients after ACLR with bone patella-tendon bone graft (305 patients) had an average knee flexor LSI of 97% 6 months after ACLR. Two-thirds of patients had ≥ 90%, whilst ACLR patients with HG had an LSI of 89%, with less than half (46%) achieving the 90% LSI cut-off. Recent research assessing knee flexor strength isometrically at 4, 8 and 12 weeks after ACLR with HG reported a LSI of 54, 70 and 76% [19]. Knee flexor deficits after ACLR are also typically more pronounced at shorter muscle lengths/greater knee flexion angles [59], slower velocities [59] and for rate of force development (RFD), than maximal strength [60, 61]. One study has assessed knee flexor RFD after ACLR and reported a lower LSI for RFD during knee flexion (and extension) (average 55%) than maximal force (66%). There was though a higher RFD knee flexor to extensor ratio (average 0.63) compared to the contralateral limb (0.44), due to large deficits in knee extensor RFD [61].

The hamstring muscles are responsible for more than just knee flexion, contributing to hip extension, as well as knee and pelvis stability. In particular, the medial hamstrings are thought to be important for preventing ACL injuries, due to their role in preventing medial condyle lift-off and dynamic knee valgus [17], a known ACL injury risk factor [62, 63]. Of particular relevance after ACLR with HG, is the commonly observed deficits in knee internal rotation strength weakness [64, 65], which would be expected to contribute to the increased external tibial rotation and dynamic knee valgus found in ACLR patients [66].

Muscle architectural changes can also occur independent of muscle size [67] and the biceps femoris long head (BF_{LH}) of the ACLR limb has been shown to demonstrate shorter muscle fascicles and greater pennation angles after ACLR with HG [24]. The architectural changes in BF_{LH} are comparable to the changes observed in those with previously HSI of BF_{LH} [68]. Recent evidence suggests that professional soccer players with shorter BF_{LH} fascicles (< 10.56 cm) were four times more likely to sustain a future HSI than those with longer fascicles, and that the probability of injury was reduced by around 20% for every 1 cm increase in fascicle length [68]. It has been hypothesised that possessing shorter muscle fascicles, with fewer in-series sarcomeres, may result in an increased susceptibility to eccentrically-induced muscle damage [68, 69], which may predispose the athlete to increased running related HSI.

2.2 Why are There Residual Deficits After ACLR?

There are multiple reasons why residual deficits in hamstring function may be apparent after ACLR. Understanding these reasons can aid in better design of interventions to address deficits. The deficits will in part depend on the function of the person pre-surgery, rehabilitation factors and the specific surgical technique/graft used.

First, any deficits in hamstring function pre-surgery may be present or magnified post-surgery. Knee flexor weakness is a risk factor for primary ACL rupture [70] as, such predisposing weakness may have played a role in the injury. It is important to consider not only relative strength deficits (e.g., LSI), but also absolute strength values during the functional recovery process and to support RTS decision-making.

Second, residual deficits after ACLR, can also be in part due to an insufficient functional recovery process [56], or due to poor compliance after ACLR [71]. A key goal of the functional recovery process is to restore these deficits [9]. Reasons for the on-going deficits have in part been described due to insufficient volume, intensity and frequency of rehabilitation to target the deficits [8, 9, 56].

Finally, more marked deficits in knee flexor function are apparent after ACLR with HG, thought due to issues associated with the donor site. ACLR with HG, essentially results in a grade 4 muscle–tendon lesion. It is well aware after serious muscle lesion, there is often persistent hamstring strength deficits and high risk of HSI re-injury [55, 72, 73], partly explained by chronic neuromuscular inhibition [74]. This may result in a reduced capacity to voluntarily activate the hamstring muscle during eccentric, but not concentric knee flexor efforts [75, 76]. After ACLR with HG, there also appears to be a proximal migration of the semitendinosus (ST) muscle–tendon junction [77, 78]. ST tendon regeneration may take approximately 18 months [79] and may not occur at all in 10–50% of patients [21, 64, 78, 80]. Rehabilitation during this time and for individuals with no tendon regeneration would presumably not load the ST significantly. Muscle size and activation deficits may relate to tendon regeneration, with deficits in ST and gracilis muscle size been greater for tendons that did not regenerate [80].

Typically, there is selective ST muscle atrophy (10- 28%%) with HG [25, 78, 81, 82], commonly accompanied by gracilis muscle atrophy (~30%) [81, 82]. If these deficits are not compensated for with hamstring hypertrophy (e.g. BF and semimembranosus [SM]), this will lead to total hamstring volume deficits, which would compromise sagittal plane force capabilities (i.e., knee flexion strength). Additionally, without compensatory SM muscle hypertrophy, there would be residual deficits in medial to lateral hamstring muscle volume, which would directly influence transverse plane knee control. Messer et al. [26] reported compensatory hypertrophy of the SM after ACLR with HG, but found 18% deficit in medial hamstring volume, due to large deficits in ST volume (30%). Other studies have also reported compensatory hypertrophy of BF and not SM [80], which would exacerbate the lateral to medial muscle size (and strength) imbalance.

3 Restoring Hamstring Function After ACLR: Key Considerations in Programme Design

The majority of the research on hamstrings is focused on hamstring function for injury prevention and/or rehabilitation of HSI. There is a lack of actual evidence or advice for recovery of hamstring function in ACLR patients. In the absence of specific evidence on ACLR, it is important to consider, learn from and translate the exiting knowledge on hamstring rehabilitation and prevention to the ACLR athlete. Effective hamstring strength recovery is embedded in (1) understanding how to train the hamstrings; (2) understanding the deficits after ACLR and tailoring the hamstring conditioning approach to the required adaptive strategy; (3) understanding the ACLR journey and how to directly apply the approach to the ACLR patient. Section 2 considered the deficits after ACLR and this section will consider how to train the hamstrings after ACLR, considering effective training principles, but also the specifics of an ACLR patient, to understand how these principles need to be adapted. This will cumulate in Sect. 4 where specific recommendations for each stage of the functional recovery process, and specific progression criteria will be presented.

3.1 Achieve Optimal Loading

The cornerstone of effective functional recovery is achieving 'optimal loading', (defined as the load applied to structures that maximises physiological adaptation [83]) to bring about specific neural, morphological and mechanical adaptations [83]. A significant challenge for rehabilitation specialists is designing optimal training programmes that facilitate training adaptations, whilst been mindful of biological healing constraints, and tissue capacity [84, 85]. Traditionally, training recommendations are based on evidence in un-injured athletes, such as the American College of Sports Medicine [86]. There is no consensus on specific training programme recommendations for injured athletes. The optimal loading will depend on the desired mechanical (e.g., maximal strength, RFD, power, muscle strength endurance), neural (intermuscular coordination, motor unit recruitment/ firing frequency) or morphological (e.g., muscle size/volume, architecture, composition, tendon unit properties) adaptations. Various programming principles, such as exercise intensity, volume, time under tension and rest between exercise sets, are important in terms of achieving the optimal adaptation.

Load refers to the amount of weight assigned to an exercise set and is probably the most important variable in resistance training programme design [87, 88]. Strength gains can be observed across many loading intensities, as long as a minimum intensity is achieved, thought to be around 40–60% voluntary activation [89]. Greater gains in strength, likely due to superior neural adaptation's occur with higher intensities [90], with a dose–response relationship between intensity and strength gains [91–93]. Heavy resistances also appear to be essential for promoting adaptions in RFD and maximal eccentric strength [94–97]. Lighter resistances to fatigue (e.g., sustained efforts/high repetitions, with minimal recovery between sets) are effective for training muscle endurance/work capacity. Intensity is less essential for muscle hypertrophy, likely due to the differing mechanisms available to induce muscle hypertrophy (e.g., mechanical tension, muscle damage and metabolic stimuli) [98].

Training volume appears essential for promoting gains in muscle hypertrophy [99]. It is thought that high-volume training may enhance muscle mass gains due to prolonged metabolic stress [100]. High-volume training appears less important for promoting maximal strength or eliciting architectural adaptations [101].

To fully restore neuromuscular performance after ACLR, incorporation of a periodised neuromuscular training programme, respecting tissue healing times and the patients individualised functional recovery, appears important. Periodisation can be defined as the planned manipulation of training variables (load, sets and repetition) to maximise training adaptations and prevent over-training [84, 102, 103]. Knee flexor strengthening in ACLR patients with HG is normally suggested to be delayed for 6–8 weeks after surgery to allow healing of the harvested graft [104–106]. However, there is no strong evidence for this and lower intensity exercises are advised earlier after surgery, based-on anecdotal experience. Those without hamstring graft can be less cautious, respecting the load capacity of the knee as a whole. Initially, lower intensity loading should be used during the earlier stages after ACLR (i.e., end of early-stage to mid-stage), when the knee is load compromised and there is likely accompanying swelling and pain, as well as arthrogenic muscle

inhibition (AMI), thus contraindicating high loads [9]. This initial lower intensity training will target the recovery of muscle endurance and work capacity as well as hypertrophy through metabolic adaptations [102]. This may be supported by supplementary modalities, such as blood flow restriction training [102, 107, 108]. As an athlete gets stronger and overcomes pain, swelling and AMI, higher intensities can and should be used, in a progressive manner. This should initially focus on muscle hypertrophy, with higher volumes at moderate loads (e.g., 6–8 sets of 8–12 RM). This period can then be followed by a greater focus on maximal strength (and improvements in neuromuscular activation), with higher resistance loads and moderate volume. Finally, a transition to explosive strength and power training, in conjunction with very high-intensity, low-volume resistance training (e.g., < 5RM) is recommended in the final stages prior to RTS. See Table 1 for general programming guidelines for each mechanical variable.

3.2 Take Some Lessons from Hamstring Rehabilitation Prevention and Incorporate 'Eccentrics' and Longer Length Strengthening Exercises

Contraction type is the most commonly monitored parameter in strength training to prevent HSIs [87]. It is though less frequently discussed when considering hamstring function after ACLR, which typically considers isometric and/or concentric strength via either isometric or isokinetic testing [28, 29, 109, 110]. Adaptations to strength training are mode-specific [111–113]. Given that ACL graft failure likely occurs when the hamstring muscle actively lengthens to resist anterior tibial translation [114], and that HSIs mainly occur when the hamstrings act eccentrically to brake the knee extension at the end of the running swing phase [115], it would seem relevant to prioritise eccentric strengthening prior to RTS. In the little research published, it appears that eccentric hamstring deficits after ACLR with HG are ~ 16–20% [22, 23, 25]. In terms of HSI, it appears that higher levels of eccentric but not concentric knee flexor strength have been shown in most [116–119], but not all prospective studies [116, 120] to be associated with a reduced risk of HSI.

Eccentric training overloads the muscle to a greater extent and enhances muscle mass, strength and power more than concentric training [121]. Improvement in knee flexor eccentric strength after 6–10 weeks of knee-based eccentric hamstring strengthening is typically 13–19% [111, 122, 123]. Eccentric training in general has also been shown to result in a rightward shift in the torque–joint angle relationship of the knee flexors [124–127], thought due to alterations in muscle fascicle length, which has been shown to increase after eccentric, but not concentrically based resistance training [127]. In particular, a 16–34% increase BF_{LH} fascicle length following 6–8 weeks of eccentric knee flexor training was reported [118, 127], whilst in one of these studies [118], the authors reported a 6% shortening of BF_{LH} fascicles following concentric only training on the same device. Both low [128, 129] and high-volume [129–132] programmes employing the eccentric-only Nordic hamstring exercise (NHE) observed a 13–24% increase in BF_{LH} fascicle length across a 4- to 10-week training period.

Alterations between isometric and concentric actions versus eccentric actions can occur in part due to altered muscle morphology, but more specifically due to altered neuromuscular activation during eccentric contractions [133]. It is thought after HSI and possibly ACLR with HG, that there is inhibition of the hamstring muscles (specifically the donor/injury site) which results in lower neuromuscular activation during eccentric actions and causes residual deficits in eccentric strength and muscle morphology [26, 74]. As such, although eccentric training may be highly effective, failure to overcome issues associated with hamstring AMI after ACLR, particularly with HG would limit their effectiveness. Thus, eccentric training for the ACLR needs to be considered within the overall functional recovery process.

Evidence suggests that training at longer muscle lengths can achieve similar adaptations in muscle fascicle length to eccentric strengthening. Fascicle length changes for the vastus lateralis were similar after 10 weeks of concentric versus eccentric training at long muscle length [134], whilst

Table 1 A general overview of the goals of training, the intensity and repetition ranges to optimise the training goal, the typical volumes and rest periods as well as the typical mechanisms of enhancement in function

Goal	Intensity	Rep range	Volume	Rest period	Mechanism
RFD/Power	Maximal	1–5	Low	Long > 2 min	Rapid neural activation and explosiveness of muscle tendon unit
Strength	Moderate-high	2–8	Moderate	Moderate-2 min	Maximal activation and/or muscle hypertrophy
Hypertrophy	Low to high (but typically moderate)	8–15	Moderate-high	Short-moderate (30–60 s)	Hypertrophy through -Metabolic -Mechanical tension and/or -Muscle damage
Muscle endurance	Low-to moderate	15 +	High	Short—30 s	Muscle specific metabolic adaptations

10 weeks of conventional (combined eccentric and concentric contractions) hip extension training at long hamstring lengths resulted in a 13% increase in BF_{LH} fascicle length [131]. Thus, longer length isometrics and concentrics could precede high-intensity eccentrics to elicit positive architectural adaptations.

3.3 Balance the Use of Knee and Hip-Dominant Exercises

Although following a knee injury, one may consider hamstring function about the knee, i.e. 'knee flexor strength' to be more important, it is also important to consider hamstring function in conjunction with other muscles (i.e. gluteus maximus, adductor magnus) about the hip, i.e. 'hip extension strength'. Knee flexor weakness is a risk factor for both ACL [29] and future HSI [55, 135]. Interventions aimed at increasing knee flexor strength, particularly eccentric knee flexor strength, have reduced HSI rates, across multiple sports [136–139] and so of course developing knee flexor strength is essential. However, during certain functional tasks, such as the swing phase of sprinting, the moment arm and internal moments at the hip are double that at the knee [140, 141] and the fascicle length of the hamstring muscles (BF mainly) are more sensitive to hip position [142, 143]. Weakness in hip extension strength was identified as a prospective risk factor for HSI in elite level sprinters [144]. Furthermore, weakness of hip extensors may contribute to altered motor patterning in the sagittal plane, leading to increased knee loading motor patterns (Fig. 1) [145]. Thus, it is important to consider both knee flexor and hip extensor strength.

As we know, skeletal muscle activation has the potential to influence the functional and structural adaptations to resistance training [131, 146, 147]. There is evidence to suggest that the hamstrings are activated heterogeneously during a range of different exercises [148–153]. Knee dominant exercises (e.g. prone leg curl [150] and NHE [148, 149, 154, 155]) are thought to preferentially activate, as well as result in specific adaptations to chronic exposure to the ST and BF short head. Hip extension exercises (e.g., the stiff-leg deadlift [151] or 45° hip extension exercise [131]) appear to involve to a greater extent and result in more specific adaptations of the SM and BF_{LH}, as well as at the more

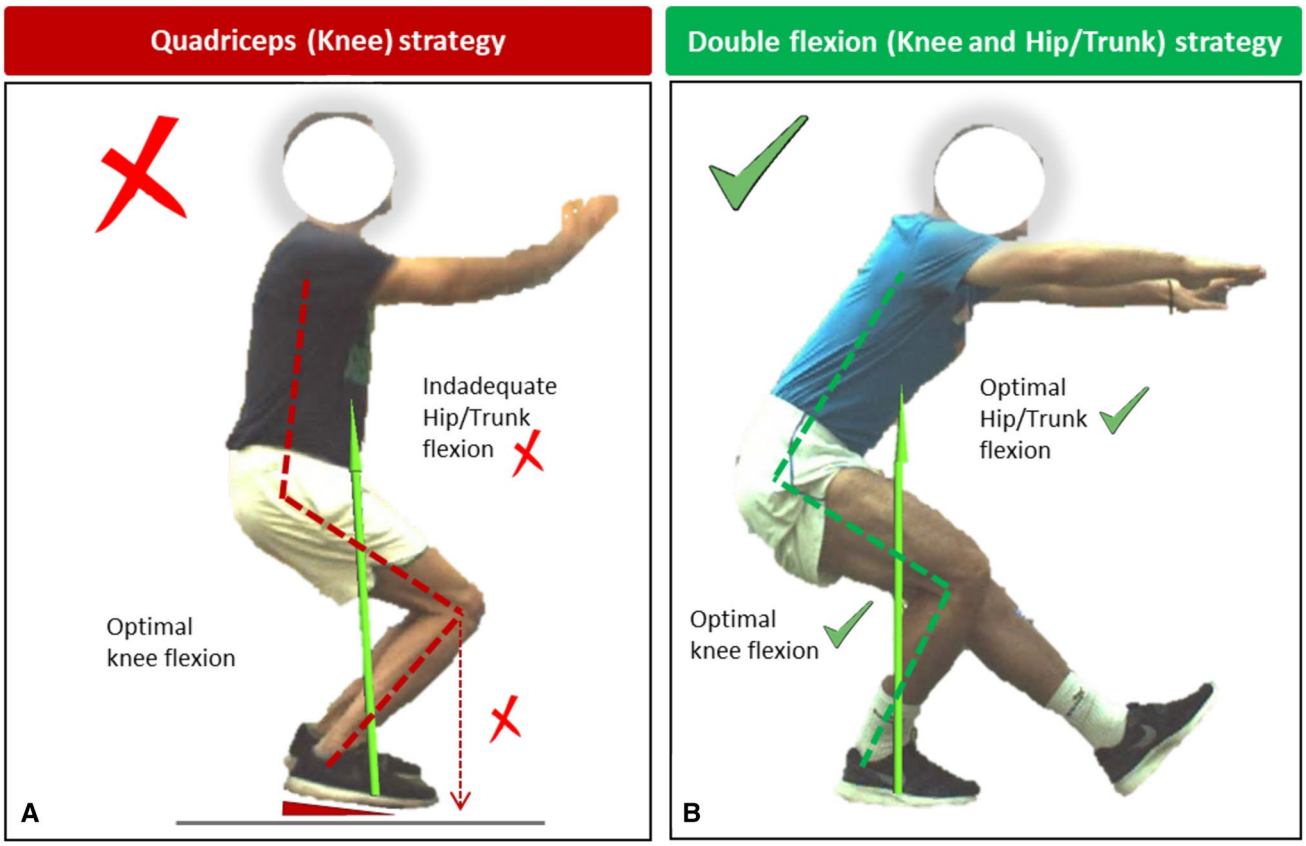

Fig. 1 An example of a knee/quadriceps dominant movement strategy with upright trunk, resulting in greater knee load. This is most commonly associated with the knee excessively positioned anterior to toes (**a**) An optimal movement strategy balancing hip and knee contributions, with the knee slightly but not excessively over the toe and similar hip and knee flexions (**b**)

proximal regions of the muscle. This reiterates the need for the use of a balanced approach of both knee flexion and hip extension dominant exercises, to target all the hamstring musculature (Fig. 2). The specific neuromuscular activation within and between muscles can be modulated during most of these exercises using certain techniques. One example, been changing foot rotation position, which has been shown to selectively upregulate the hamstring muscles, with foot internal rotation increasing activation of the medial hamstrings and foot external rotation increasing the activation of lateral hamstrings [156].

A key consideration after ACLR with HG is whether or not to target the ST. Strong use of knee dominant exercises, indicative of ST specific training, with failed ST tendon regeneration may result in overcompensation of BF short head and altered rotational control about the knee. In this case, targeting the SM with hip-dominant exercises to compensate for the ST tendon issues and maintain medial to lateral hamstring muscle balance, may be the superior strategy. However, in the case of ST tendon regeneration, but accompanying AMI, there would appear a need to include specific activation/strengthening exercises for the ST, to provide sufficient stimulus for muscle adaptation. In this case, it is suggested to adopt strong focus on knee dominant exercises using a periodised approach. Actual evidence on this topic is though missing.

3.4 Think About Functional and Not Just Isolated Strength

Strength is the ability to produce force and is influenced by morphological, biomechanical and neural factors; the contributions of each depend on the strength task [157]. Functional strength is the ability to produce force in situations in which the muscles are commonly used (and injured) [157], whereas isolated strength tasks minimise the degrees of movement freedom and isolate the muscle group of choice, and as such do not mimic the way in which the muscle's function [157]. As described, the hamstring muscles must function during complex tasks to prevent injury, typically eccentrically during acceleration, deceleration and sprinting tasks. These tasks typically require high degrees of fine motor control. There are numerous mechanical and neural differences between isolated force development and force production during these complex sporting tasks. Although isolated strength serves as a 'capacity' to produce force, certain factors may limit the transfer of isolated strength to functional situations.

First, Sale [158] likened the expression of strength to a skilled act where agonists must be maximally activated, whilst supported by appropriate synergist and stabiliser activation and opposed by minimal antagonist co-activation. Poor intermuscular coordination, specifically lower than optimal stabiliser muscle recruitment can result in

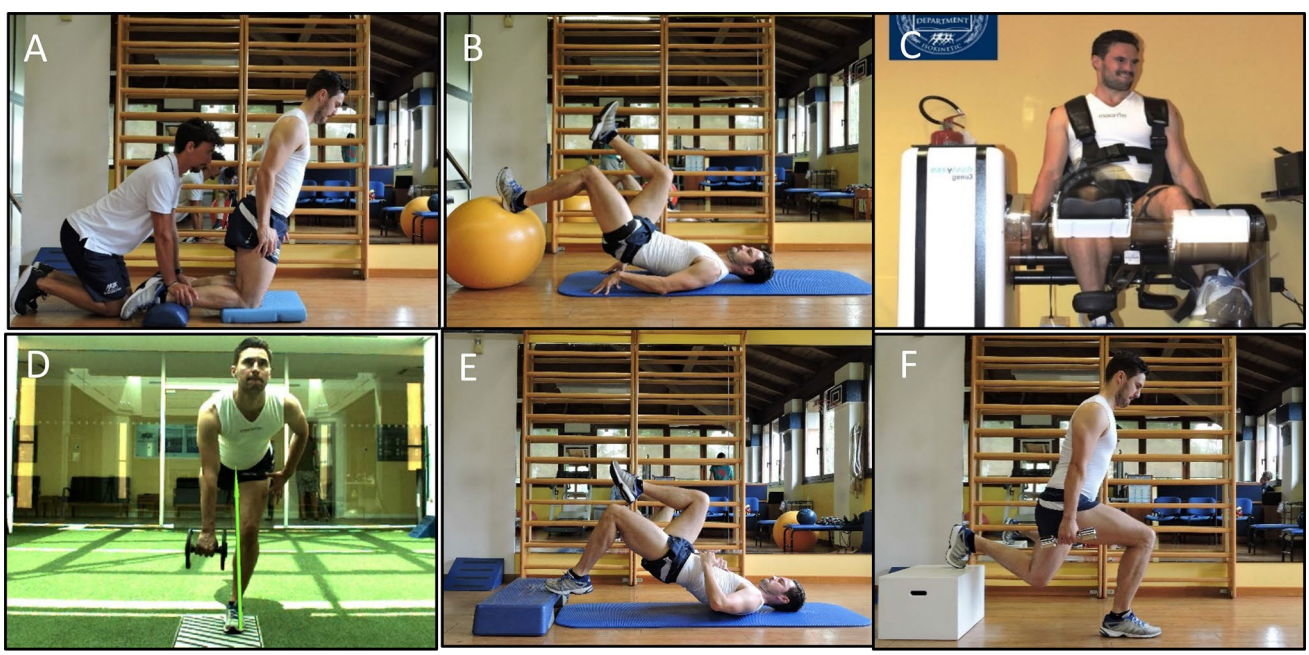

Fig. 2 Examples of knee dominant hamstring exercises, **a** Nordic hamstring exercise, **b** Swiss ball single leg roll outs and **c** isokinetic knee flexion as well as hip-dominant hamstring exercises, **d** single-leg stiff-leg loaded deadlift, **e** single leg bridge on box (<90° knee flexion) and **f** rear foot-loaded elevated split squat (as an example of weight-bearing functional exercises, not specifically hamstring dominant exercises)

insufficient expression of isolated strength functionally due to agonist and antagonist compensation for dynamic stability, thereby compromising force output [159, 160]. The stabiliser muscle system is becoming recognised as an important contributor to functional strength [160]. There is also evidence that weakness of certain muscles in the lumbo-pelvic-hip or 'core' region may increase the risk of lower limb and/ or ACL [145, 161–166] and hamstring injuries [48, 167, 168]. Core stability training has been shown to result in reduced hamstring stiffness, likely due to reduced requirements for the hamstring muscles to compensate as global stabilisers [167]. As well as core 'stability', altered 'pelvic balance' may also contribute to reduced hamstring muscle performance. For example, increased hip flexor tension and reduced hip flexor range of motion (potentially secondary to decreased anterior pelvic stability and compensation) results in both an anterior tilted pelvis and altered intermuscular coordination. Reduced hip flexor range of motion has been associated with reciprocal inhibition of gluteus maximus and synergistic dominance of the hamstrings [169]. Furthermore, an anterior tilted pelvis elicits a stretch on the hamstring muscle resulting in altered length–tension relationship, thereby reducing the ability of the muscle to produce force at longer muscle lengths, typically indicative of injury. This situation would be exacerbated by weakness in hip extension, as well as sacroiliac joint dysfunction [170].

Residual movement impairments are apparent during an array of functional tasks in ACLR patients [171–176]. ACL injury and subsequent reconstruction can lead to altered movement quality of patients bilaterally [173] and has been prospectively linked with elevated secondary ACL injury risk [177, 178]. As such, movement re-training prior to RTS should form an important element of the functional recovery process [8, 9]. Altered movement quality is thought to be due to multiple factors including altered posture (e.g. anterior tilted pelvis, [179, 180]), arthrokinetic dysfunction (e.g. reduced dorsi-flexor range of motion, [181]), muscle imbalances/weakness (e.g. knee extensor weakness, [182]), altered reciprocal muscle inhibition and synergistic dominance [169, 183] and altered proprioception [184, 185]. The hamstring muscles play an important role in supporting optimal lower limb control and alignment [17]. In particular, the medial hamstrings are considered important for preventing ACL injury, due to limiting dynamic knee valgus [17], a key movement pattern associated with ACL injury [62, 63].

Alterations in hamstring activation and control have been shown to lead to increased ACL risk. Zebis et al. [186] prospectively linked altered neuromuscular pre-activation of the thigh muscles during a side step cut to ACL injury risk in a cohort of volleyball players during the subsequent 2 years after screening. In particular, a reduced ST to vastus lateralis neuromuscular pre-activation ratio (a 50 ms window prior to ground contact) was associated with elevated ACL injury risk. Muscle pre-activation and feedforward motor strategies are an overlooked but essential aspect of neuromuscular function. ACL ruptures have been shown to occur within 50 ms after ground contact [187]. This is about twice as long as the ACL/ hamstring reflex arc [188]. Thus, the reflex is longer than the injury and therefore, injury prevention in this situation does not rely on feedback processes. Therefore, dynamic stabilisation of joints via muscular pre-activation is essential for joint injury prevention. During movement there is pre-activation of the muscles with recorded surface electromyography (sEMG) values around 125–150 ms before ground contact [186, 189], which allows for the development of tension and thus active stabilisation of joints prior to landing. These sEMG values have been reported to be around 40–60% of maximal sEMG prior to ground contact (10–50 ms time window before foot contact on force platform). A key aspect of hamstring reconditioning for ACL re-injury prevention appears achieving optimal motor patterning during sporting-type tasks. The role of dynamic knee valgus in ACL injury mechanisms [190–192] suggests that optimal medial hamstring function, particularly pre-activation is important in preventing ACL injuries [17, 62, 63]. Utilising neuromuscular re-training with motor pattern re-learning during sporting-type movements to enhance both kinetics (strength/neuromuscular capacity) and kinematics is therefore recommended [8]. Neuromuscular training involving jumping, landing and plyometric type tasks has been shown to both reduce the rate of ACL injuries by half [193] and additionally result in selective upregulated of the medial hamstring muscles [189].

The ability to develop force rapidly, as well as at high muscle speeds is important to produce force during rapid sporting movements. Mechanical stabilisation of the joint to prevent injury, as well as explosive movements, such as sprinting, involves ground contact times considerably shorter than the time to produce maximal force (50–150 ms vs. 250–300 ms) [194–196]. Additionally, between ~ 25 and 80% of the sprint running cycle, the hip is flexing with a peak velocity greater than 700°/s [197], whilst between ~ 55 and 95% of the sprinting cycle, the knee is extending with a peak angular velocity greater than 1,000°/s [189]. Deficits in RFD or high-speed strength would be expected to compromise hamstring performance under explosive functional tasks. Thus, it is important to consider the hamstrings 'explosive' neuromuscular performance and the ability to generate force rapidly and the associated ability to increase neuromuscular activation from low to high levels [97, 198]. In terms of explosive strength/activation, conventional resistance training, such as sustained high force contractions or isoinertial resistance training (even with maximal intention to lift the weight during the concentric phase), has been shown to exert minimal benefits on RFD [96, 97, 159] and no change in early phase neuromuscular activation [97,

159]. This is thought because in response to sustained contractions, the neuromuscular adaptations are specific to the high force aspect of the force–time curve [97, 159, 194]. It appears that for gains in RFD and explosive neuromuscular activation, there is a need for exercises which have rapid increases in force and activation during the task [199]. Most isolated and weight-bearing functional tasks involve sustained levels of activation and little explosive element to them. As such, there is a need for some additional tasks to challenge this aspect of neuromuscular function.

We believe a key consideration of effective neuromuscular training is to balance the use of both isolated and functional exercises to enhance muscle strength, address factors influencing movement quality and optimise muscle coordination and motor patterning during foundation and sporting-type tasks [9]. Inclusion of corrective lumbopelvic-hip training and functional neuromuscular strength exercises, such as foundational tasks (e.g., single leg squat/stiff-leg deadlift), landing and ballistic drills, plyometrics and agility training, is recommended alongside specific isolated hamstring strengthening, as part of a holistic approach to hamstring re-conditioning [157, 200]. It is known that enhanced muscle strength does not directly transfer to enhanced functional performance (kinetics and kinematics) [201–203]. Instead, coordinative changes are required to be able make full use of the enhanced muscle strength [201]. Functional strength exercises require greater coordination and result in task-specific adaptations, due to neural adaptations [159]. They can also support the enhancement of isolated strength, providing sufficient neuromuscular activation and work volumes [160]. Furthermore, tasks, such as landing, plyometric tasks and sporting-specific movements, will challenge inter- and intramuscular coordination, and when performed with appropriate technique and biofeedback aid in optimising coordination and patterning during these tasks [204]. Regaining symmetry in high-load sporting tasks may be associated with lower re-injury risk [205].

A weakness of most functional tasks is that they achieve relatively low levels of neuromuscular activation of the hamstring musculature versus other more isolated exercises [148]. So, they will not provide the necessary stimulus for the necessary neuromuscular adaptations. Although isolated strength tasks, such as the seated leg curls or NHE (knee dominant) or 45º hip extension/bridge exercises (hip-dominant) lack specificity and transference to functional exercises/movement [202], they are considered important for hamstring strengthening. They effectively remove the need for voluntary stabilisation of joints and/or synergistic control of force direction (and thus the need for optimal synergist and/or stabiliser muscle recruitment) and are excellent to target a specific muscle group in isolation. Therefore, they can be highly effective in the presence of specific muscle weakness or inhibition, or when load compromised during the early periods after ACLR. Thus, we recommend a combination both strength and neuromuscular training exercises to optimises neuromuscular function and motor control [206].

4 Summary and Recommendations for Implementation: Functional Recovery of Hamstrings Over the ACLR Programme

Although there is no consensus on the functional recovery process after ACLR, recent work has been devoted to standardising the ACL rehabilitation journey. Current best practice for ACL rehabilitation appears to involve criterion-based rehabilitation through a series of stages [8, 9, 207]. The functional recovery process can be broadly separated into pre-operative, early-, mid- and late-stage rehabilitation and RTS training (Fig. 3) [8, 9]. It is important that the hamstring strengthening approach is placed in context of the overall goals of the programme. These include restoring knee function (e.g., pain, swelling, range of motion) as well as neuromuscular function (e.g., maximal isolated and functional strength and explosive neuromuscular performance) of many muscle groups of the lower limbs. Hamstring re-conditioning after ACLR is not as simple as using a series of progressive hamstring exercises, but instead recognising the health status of the athlete, understanding how to train the hamstrings, and knowing how this hamstring conditioning approach needs to be adapted to the ACLR athlete depending on their specific level of functionality.

In general, we recommend a holistic approach to hamstring re-conditioning [157, 200], after ACLR. This approach should focus on all aspects of function (e.g., neural and morphological and mechanical factors) and respect the unique neuromuscular benefits of different exercises. There are differing levels of functional tasks, including weight-bearing, high-load control (landing, jumping/plyometric) and sport-type/specific (speed, change of direction

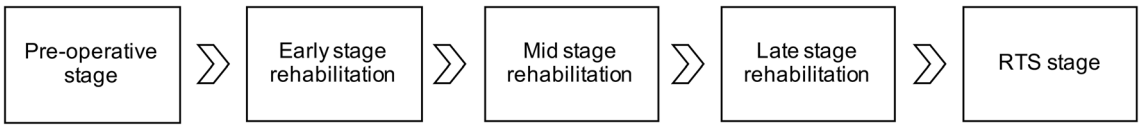

Fig. 3 The functional recovery processes after anterior cruciate ligament reconstruction. Modified from Buckthorpe and Della Villa [9]

and agility) tasks (Fig. 4) and they can be characterised according to the level of intensity and specificity. In terms of intensity, it is important to consider both the external joint loading (e.g., peak ground reaction forces, rate of force acceptance/production and volume loading), as well as the internal joint torque/moments and muscle activation. Each task may have unique benefits at a particular time of the recovery process, and so knowing when to use which task and in which way is a key aspect of the 'art' of rehabilitation/re-conditioning. Table 2 contains our recommendations on hamstring conditioning after ACLR, using a criterion-based approach, with specific stage goals and hamstring goals after ACLR. There is a progressive transition from low to high load, static to fast dynamic, isolated to functional, and isometric to eccentric. Below we provide suggestions for hamstring functional recovery after ACLR with HG across the progressive stages, tailored for ACLR with HG.

4.1 Early-Stage

Early-stage rehabilitation is focused on resolving pain and swelling, recovering sufficient knee joint range of motion, recovery of activities of daily living including the ability to walk without crutches, and minimisation of muscle atrophy [9, 208]. Deficits in knee flexor isometric force of nearly 50% have been reported at 4 weeks after ACLR [19], which generally coincides with a transition from early- to mid-stage rehabilitation [9]. Minimising knee flexor strength deficits during the early-stage would make strength recovery much easier during the subsequent mid-stage stage of rehabilitation. Current recommendations are to delay specific strengthening of the hamstrings after ACLR with HG for 6–8 weeks [104–106] to facilitate appropriate donor site recovery. However, we advise use of isometric/concentric exercises of low intensity at short-medium muscle lengths. It is important during this stage though to avoid strenuous activities which may potentially result in damage to the graft and/or donor site.

4.2 Mid-Stage

The mid-stage has been described as having three primary goals, (1) resolution of large muscle strength asymmetries; (2) restoration of movement quality during foundation motor tasks (weight-bearing functional tasks and jogging on treadmill) and physical fitness re-conditioning [9]. In terms of hamstring rehabilitation, the mid-stage aims to adopt a relatively simple programme to overcome large (typically 50%) [19], knee flexor strength deficits on a load compromised knee. A key goal in mid-stage, is to restore knee flexor maximal strength to within at least 20% of the contralateral limb [8, 9]. This is typically done alongside some basic motor patterning restoration (e.g., functional exercises, such as squat and running gait) [9], which will facilitate appropriate motor patterning on which to utilise a stronger functional strengthening approach during the late-stage. Recently, we discussed how to optimise the mid-stage after ACLR in general [9] and suggested separating mid-stage into a first and second half. The first half in regards to hamstring conditioning would use low-to moderate loads, potentially conjunction with supplementary modalities, such as blood flow restriction training to restore muscle volume and work capacity [9]. A balance of both knee and hip-dominant exercises is recommended, with a focus on isometric or concentric actions. Progression to the second half is based on clinical experience as opposed to specific criteria [9], but involves a transition to

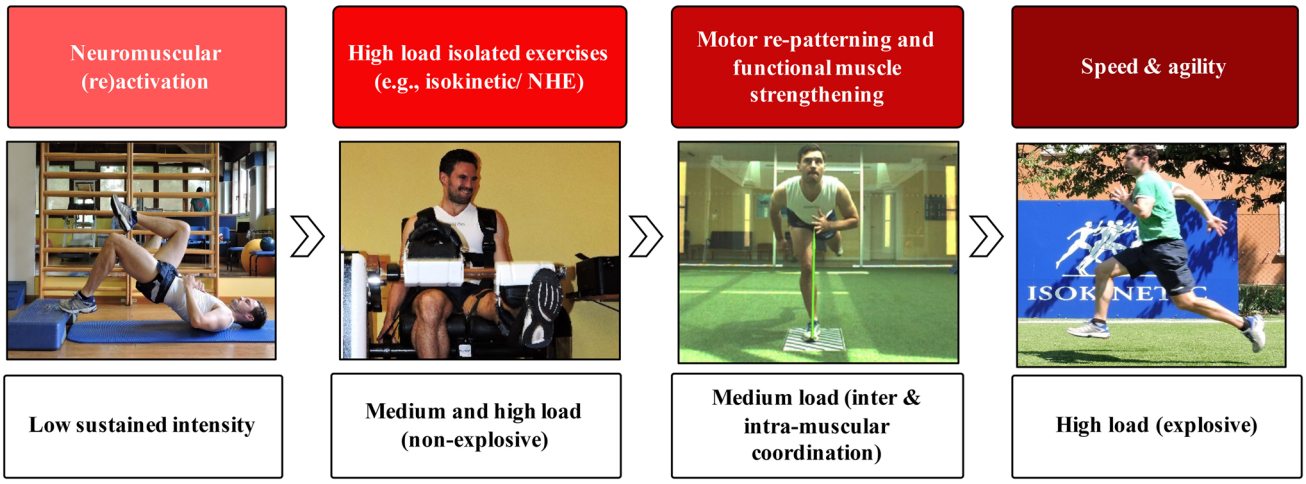

Fig. 4 The four exercise categories to support exercise selection after anterior cruciate ligament reconstruction including neuromuscular re-activation exercises, high-load isolated strength exercises, foundation and high-load (jumping/landing) motor re-patterning and speed and agility exercises. Each exercise will have differing loading properties and stimuli for neuromuscular adaptation

Table 2 Hamstring exercises according to the functional recovery stage after ACL reconstruction, with suggested time lines, goals of the stages, specific hamstring goals, training planning considerations and specific exercises

Stage	Early-stage	Mid-stage	Late-stage	RTS training
Typical weeks	0–4	5–12	13–18	19–24
General goals of stage	Resolution of pain and swelling Recovery of knee joint ROM Recovery of normal walking gait without aid Preservation of muscle size (particularly quadriceps)	Full recovery of joint ROM Restoration of muscle strength imbalances to within 20% of 'trained' contralateral limb (or pre-injury strength values) Recovery of basic motor patterning and running gait Avoid physical fitness detraining	Restore neuromuscular function markers to within 10% (knee and adjacent joint specific strength and closed kinetic chain and power) Restore high load movement quality Restore aerobic fitness	Restore sports specific movement quality, fitness, skills and develop movement 'volumes' to prepare for RTS
Hamstring specific goals and criteria	Recovery of donor site and/or prevent muscle atrophy and strength loss	Recovery of knee flexion maximal strength to within 20% of contralateral limb	Recovery of maximal knee flexion and hip extensor strength (LSI > 90%) Knee flexor RFD symmetry (LSI > 85%) Progress linear running speeds to 95% max speeds Progress pre-planned coordination exercises to 95%	Recover maximal running speeds Restore maximal eccentric strength at longer muscle lengths Support muscle fascicle elongation
Training planning				
Intensity	Low	Moderate	High	Very high
Volume	Low	Moderate—high	Moderate	Low—moderate
Contraction type	Isometric > dynamic	Concentric/isometric > eccentric	Isometric = concentric = eccentric	Eccentric > concentric/isometric
Length	Short > > > long	Short > long	Long > = short	Long > > > short
Hip vs knee	Hip > knee	Knee = hip	Knee = hip	Knee = hip
Functionality	Isolated	Isolated > functional	Functional > isolated	Functional > > isolated
Velocity	Static	Slow	Slow = fast	Fast > slow
Explosivity	Low	Low > moderate	Moderate—high	High—very high
Example hamstring exercises	Neuromuscular activation exercises and low intensity-controlled movements such as co-contractions, controlled prone limb lowering, controlled knee flexion recovery in gait (pool priority)	1st half of programme in HG: Seated knee flexion isometric and isokinetic Moderate load hip extension exercises Prone manual strengthening isometric and sub-maximal dynamic	Straight leg load bridges Nordic hamstring exercise Good morning/ RDL (BL and UL) Straight line runs of increasing speeds Multidirectional runs/curved and CoD	Straight leg load bridges Nordic hamstring Good morning/ RDL (BL and SL) Straight line runs of increasing speeds Agility training Sport-specific practice

ROM range of motion, *CoD* change of direction, *RTS* return-to-sport, *LSI* limb symmetry index, *HTG* hamstring tendon autograft, *RDL* Romanian deadlift, *BL* bilateral, *UL* unilateral

both isolated and functional exercises at moderate loads (6–12 RM), to maximise muscle volume and reduce muscle strength deficits. Emphasising optimal technique is essential during functional exercises to ensure positive motor control adaptations [9, 204].

4.3 Late-Stage Rehabilitation and RTS Training

The late-stage rehabilitation and RTS programme involves a stronger emphasis on hamstring functional recovery. Recent work by Buckthorpe [8] provides a framework for late-stage rehabilitation and RTS training involving 5 progressive stages. In general, late-stage rehabilitation focuses on optimising neuromuscular and movement performance, and RTS training, defined as a continuum of sport-specific on-field rehabilitation, return to training, return to play [8, 10]. In particular, the programme using a balanced approach of high-load isolated strengthening, functional strength exercises, motor pattern re-training and on-field reconditioning [9]. It aims to optimally prepare an athlete safely for RTS at low risk of re-injury and in terms of hamstring function, aims for (1) full normalisation of isolated knee flexor and hip extensor muscle strength; (2) optimisation of hamstring control during sporting-type tasks; (3) full restoration of eccentric strength at long muscle lengths and (4) restoration of maximal and explosive strength across all velocities.

Restoring explosive muscle performance is a key aspect of late-stage rehabilitation and RTS training. One training modality which has received little attention in functional recovery after ACLR but has done so more recent for HSI rehabilitation/prevention, is used of high-speed running [157, 200, 209, 210]. Running (high speed and sprinting) has both high specificity and intensity, both in terms of the resultant ground reaction forces and hamstring-specific muscle work and neuromuscular activation and may support optimisation of hamstring neuromuscular function (e.g., maximal strength, RFD, power) and motor patterning. The evidence on the use of high-speed/sprint running indicates similar gains in eccentric maximal muscle strength versus the NHE [211]. Sprint running has been shown to result in the highest neuromuscular activation of the hamstring musculature, with other exercises, such as NHE, amongst others shown to only achieve 18–75% of the EMG activation during sprinting [212]. High-speed/ sprint running demands high internal forces of the hamstrings (up to 9 Nm kg^{-1}), which is dependent on the running speed [213]. The amount of kinetic energy absorbed in the limb is proportional to the running speed squared, such that the negative work done (energy absorbed) by the hamstrings increases substantially with running speed [214]. Furthermore, the hamstring work was shown to double from 7 to 9 m s^{-1} [213]. High-speed/ sprint running is considered problematic for hamstring muscles (i.e., mechanism for injury) [136, 215] but if used appropriately considered essential for HSI prevention [200]. We suggest it forms a major element of the hamstring reconditioning during RTS, which may support a reduction in HSI reoccurrence upon RTS [54] and improve athletic/sprint performance [211].

4.4 RTS Testing

We recommend screening hamstring function after ACLR. As described, hamstring strength deficits have been prospectively shown to possess the highest risk factor for ACL re-injury after ACLR [29]. However, assessing knee flexor function does not form part of the standard ACL RTS testing, which typically involves (1) time; (2) knee function; (3) hop testing and sometimes; (4) isometric or isokinetic knee extensor testing [29, 110, 216]. Buckthorpe [8] proposed more stringent criteria for RTS after ACLR to improve the sensitivity and specificity [8]. Although, more comprehensive investigations could be made, ensuring restoration of 90% LSI knee flexor strength and 60% knee extensor/flexor ratio is recommended for all patients after ACLR. Furthermore, considering the absolute strength is recommended, with a 1.5 Nm kg^{-1} knee flexor peak torque advised prior to RTS (assessed isokinetically at 90 ° s^{-1}) (e.g., 60% of 2.5 Nm kg^{-1} for knee extensors) [8, 9]. Further assessing the knee flexor strength either isokinetically (eccentric flexor to concentric extensor peak torque > 1 and/or eccentric peak torque > 2.5 Nm kg^{-1}) or using the NordBord (LSI > 90% and peak torque > 350 N) [200, 217] is recommended where possible. Assessment of knee flexor absolute and relative (to body mass and knee extensors) RFD may be useful, but clinically challenging [8, 199]. Achieving peak running speeds during on-field rehabilitation (and quantifying with GPS technology and/or speed gates where possible) is also recommended. It is important though that there is a greater appreciation of the importance of hamstring function in ACL re-injury prevention and that sufficient time and quality of work is devoted to the recovery of hamstring function prior to RTS.

5 Conclusion

The hamstrings are important for the ACL and knee as a whole. After ACLR and more so when using the hamstrings as the choice of graft, their recovery during the functional recovery process is important. Failure to recover strength of the hamstring muscles may compromise knee health and elevate re-injury risk upon RTS. This is likely due to inability to compress and stabilise the knee and prevent alterations in frontal and traverse plane knee control, known to stress the ACL. It is suggested to emphasise greater importance on the recovery of hamstring strength after ACLR, and here is

presented some important considerations. It is recommended to utilise a holistic approach to target the recovery of both knee flexor and hip extensor strength as well as multiplanar function. This should involve a periodised resistance training programme aligned to the ACL functional recovery approach. A progressive approach from isolated low intensity to functional and high-intensity training aligned to the overall knee status is recommended with screening of hamstring function as part of criterion-based functional recovery and RTS.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of Interest The authors declare they have no conflicts of interest relevant to the content of this review.

Author Contributions MB conceived the idea for the manuscript and wrote the first versions of the manuscript. All other authors provided intellectual contribution as part of the redrafting process within their clinical areas of expertise.

References

1. Ardern CL, Webster KE, Taylor NF, et al. Return to pre-injury level of competitive sport after anterior cruciate ligament reconstruction surgery: two-thirds of patients have not returned by 12 months after surgery. Am J Sports Med. 2011;39:538–43.
2. Waldén M, Hägglund M, Magnusson H, Ekstrand J. ACL injuries in men's professional football: a 15-year prospective study on time trends and return-to-play rates reveals only 65% of players still play at the top level 3 years after ACL rupture. Br J Sports Med. 2016;50(12):744–50.
3. Lai CCH, Feller JA, Webster KE. Fifteen-year audit of anterior cruciate ligament reconstructions in the Australian Football League from 1999 to 2013: return to play and subsequent ACL injury. Am J Sports Med. 2018;46(14):3353–60.
4. Webster KE, Feller JA. Exploring the high reinjury rate in younger patients undergoing anterior cruciate ligament reconstruction. Am J Sports Med. 2016;44(11):2827–32.
5. Ardern CL, Kvist J, Webster KE. Psychological aspects of anterior cruciate ligament injuries. Oper Tech Sports Med. 2016;24(1):77–83.
6. Adams D, Logerstedt DS, Hunter-Giordano A, et al. Current concepts for anterior cruciate ligament reconstruction: a criterion-based rehabilitation progression. J Orthop Sports Phys Ther. 2012;42(7):601–14.
7. Bien DP, Dubuque TJ. Considerations for late stage ACL rehabilitation and return to sport to limit re-injury risk and maximize athletic performance. Int J Sports Phys Ther. 2015;10(2):256–71.
8. Buckthorpe M. Optimising the late-stage rehabilitation and return-to-sport training and testing process after ACL reconstruction. Sports Med. 2019;49(7):1043–58.
9. Buckthorpe M, Della VF. Optimising the 'mid-stage' training and testing process after ACL reconstruction. Sports Med. 2020;50(4):657–78.
10. Buckthorpe M, Frizziero A, Roi GS. Update on functional recovery process for the injured athlete: return to sport continuum redefined. Br J Sports Med. 2019;53(5):265–7.
11. Dingenen B, Gokeler A. Optimization of the return-to-sport paradigm after anterior cruciate ligament reconstruction: a critical step back to move forward. Sports Med. 2017;47(8):1487–500.
12. Hanson DW, Finch CF, Allegrante JP, et al. Closing the gap between injury prevention research and community safety promotion practice: revisiting the public health model. Public Health Rep. 2012;127(2):147–55.
13. Finch CF. A new framework for research leading to sports injury prevention. J Sci Med Sport. 2006;9:3–9.
14. Timpka T, Ekstrand J, Svanström L. From sports injury prevention to safety promotion in sports. Sports Med. 2006;36:733–45.
15. Verhagen E. If athletes will not adopt preventive measures, effective measures must adopt athletes. Curr Sports Med Rep. 2012;11:7–8.
16. Verhagen E, Voogt N, Bruinsma A, Finch CF. A knowledge transfer scheme to bridge the gap between science and practice: an integration of existing research frameworks into a tool for practice. Br J Sports Med. 2014;48:698–701.
17. Draganich LF, Vahey JW. An in vitro study of anterior cruciate ligament strain induced by quadriceps and hamstrings forces. J Orthop Res. 1990;8:57–63.
18. More RC, Karras BT, Neiman R, et al. Hamstrings—an anterior cruciate ligament protagonist. An in vitro study. Am J Sports Med. 1993;21:231–7.
19. Harput G, Kilinc HE, Ozer HE, et al. Quadriceps and hamstring strength recovery during early neuromuscular rehabilitation after ACL hamstring-tendon autograft reconstruction. J Sport Rehabil. 2015;24(4):398–404.
20. Ardern CL, Webster KE, Taylor NF, Feller JA. Hamstring strength recovery after hamstring tendon harvest for anterior cruciate ligament reconstruction: a comparison between graft types. Arthroscopy. 2010;26(4):462–9.
21. Nomura Y, Kuramochi R, Kukubayashi T. Evaluation of hamstring muscle strength and morphology after anterior cruciate ligament reconstruction. Scand J Med Sci Sports. 2015;25(3):301–7.
22. Tengman E, Brax Olofsson L, Stensdotter AK, et al. Anterior cruciate ligament injury after more than 20 years. II. Concentric and eccentric knee muscle strength. Scand J Med Sci Sports. 2014;24(6):e501–9.
23. Timmins RG, Bourne MN, Shield AJ, et al. Biceps femoris architecture and strength in athletes with a previous anterior cruciate ligament reconstruction. Med Sci Sports Exerc. 2016;48:337–45.
24. Vairo GL. Knee flexor strength and endurance profiles after ipsilateral hamstring tendons anterior cruciate ligament reconstruction. Arch Phys Med Rehabil. 2014;95(3):552–61.
25. Bourne MN, Bruder AM, Mentiplay BF, et al. Eccentric knee flexor weakness in elite female footballers 1–10 years following anterior cruciate ligament reconstruction. Phys Ther Sport. 2019;37:144–9.
26. Messer DJ, Shield AJ, Williams MD, et al. Hamstring muscle activation and morphology are significantly altered 1–6 years after anterior cruciate ligament reconstruction with semitendinosus graft. Knee Surg Sports Traumatol Arthrosc. 2020;28(3):433–41.
27. Kim HJ, Lee JH, Ahn SE, et al. Influence of anterior cruciate ligament tear on thigh muscle strength and hamstring-to-quadriceps ratio: a meta-analysis. PLoS ONE. 2016;11(1):e0146234.
28. Cristiani R, Mikkelsen C, Forssblad M, et al. Only one patient out of five achieves symmetrical knee function 6 months after primary anterior cruciate ligament reconstruction. Knee Surg Sports Traumatol Arthrosc. 2019;27(11):3461–70.

29. Kyristis P, Bahr R, Landreau P, et al. Likelihood of ACL graft rupture: not meeting six clinical discharge criteria before return to sport is associated with a four times greater risk of rupture. Br J Sports Med. 2016;50:946–51.
30. Verrall GM, Slavotinek JP, Barnes PG, et al. Clinical risk factors for hamstring muscle strain injury: a prospective study with correlation of injury by magnetic resonance imaging. Br J Sports Med. 2001;35(6):435–9.
31. Bittencourt NFN, Meeuwisse WH, Mendonça LD, et al. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition—narrative review and new concept. Br J Sports Med. 2016;50:1309–14.
32. Cook C. Predicting future physical injury in sports: it's a complicated dynamic system. Br J Sports Med. 2016;50:1356–7.
33. Arnason A, Sigurdsson SB, Gudmundsson A, et al. Risk factors for injuries in football. Am J Sports Med. 2004;32:5–16.
34. Freckleton G, Pizzari T. Risk factors for hamstring muscle strain injury in sport: a systematic review and meta-analysis. Br J Sports Med. 2013;47:351–8.
35. Hägglund M, Waldén M, Ekstrand J. Previous injury as a risk factor for injury in elite football: a prospective study over two consecutive seasons. Br J Sports Med. 2006;40:767–72.
36. Engebretsen AH, Myklebust G, Holme I, et al. Intrinsic risk factors for hamstring injuries among male soccer players: a prospective cohort study. Am J Sports Med. 2010;38:1147–53.
37. Gabbe BJ, Bennell KL, Finch CF. Why are older Australian football players at greater risk of hamstring injury? J Sci Med Sport. 2006;9:327–33.
38. Gabbe BJ, Bennell KL, Finch CF, et al. Predictors of hamstring injury at the elite level of Australian football. Scand J Med Sci Sports. 2006;16:7–13.
39. Orchard JW. Intrinsic and extrinsic risk factors for muscle strains in Australian football. Am J Sports Med. 2001;29:300–3.
40. Evangelidis PE, Massey GJ, Pain MT, et al. Biceps femoris aponeurosis size: a potential risk factor for strain injury? Med Sci Sports Exerc. 2015;47:1383–9.
41. Fiorentino NM, Blemker SS. Musculotendon variability influences tissue strains experienced by the biceps femoris long head muscle during high-speed running. J Biomech. 2014;47:3325–33.
42. Rehorn MR, Blemker SS. The effects of aponeurosis geometry on strain injury susceptibility explored with a 3D muscle model. J Biomech. 2010;43:2574–81.
43. Timmins RG, Bourne MN, Shield AJ, et al. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. Br J Sports Med. 2016;50:1524–35.
44. Chumanov ES, Heiderscheit BC, Thelen DG. The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of sprinting. J Biomech. 2007;40:3555–62.
45. Cummings G, Scholz JP, Barnes K. The effect of imposed leg length difference on pelvic bone symmetry. Spine. 1993;18:368–73.
46. Fousekis K, Tsepis E, Poulmedis P, et al. Intrinsic risk factors of non-contact quadriceps and hamstring strains in soccer: a prospective study of 100 professional players. Br J Sports Med. 2011;45:709–14.
47. Schuermans J, Van Tiggelen D, Palmans T, et al. Deviating running kinematics and hamstring injury susceptibility in male soccer players: cause or consequence? Gait Posture. 2017;57:270–7.
48. Sherry MA, Best TM. A comparison of 2 rehabilitation programs in the treatment of acute hamstring strains. J Orthop Sports Phys Ther. 2004;34:116–25.
49. Blanch P, Gabbett TJ. Has the athlete trained enough to return to play safely? The acute: chronic workload ratio permits clinicians to quantify a player's risk of subsequent injury. Br J Sports Med. 2016;50:471–5.
50. Bowen L, Gross AS, Gimpel M, et al. Accumulated workloads and the acute: chronic workload ratio relate to injury risk in elite youth football players. Br J Sports Med. 2017;51:452–9.
51. Duhig S, Shield AJ, Opar D, et al. Effect of high-speed running on hamstring strain injury risk. Br J Sports Med. 2016;50:1536–40.
52. Gabbett TJ. The training-injury prevention paradox: should athletes be training smarter and harder? Br J Sports Med. 2016;50:273–80.
53. Hulin BT, Gabbett TJ, Caputi P, et al. Low chronic workload and the acute: chronic workload ratio are more predictive of injury than between-match recovery time: a two-season prospective cohort study in elite rugby league players. Br J Sports Med. 2016;50:1008–12.
54. Malone S, Roe M, Doran DA, et al. High chronic training loads and exposure to bouts of maximal velocity running reduce injury risk in elite Gaelic football. J Sci Med Sport. 2017;20:250–4.
55. Opar D, Williams M, Timmins R, et al. Eccentric hamstring strength and hamstring injury risk in Australian Footballers. Med Sci Sports Exerc. 2015;47(4):857–65.
56. Welling W, Benjaminse A, Lemmink K, et al. Progressive strength training restores quadriceps and hamstring muscle strength within 7 months after ACL reconstruction in amateur male soccer players. Phys Ther Sport. 2019;9(40):10–8.
57. Hiemstra LA, Webber S, MacDonald PB, Kriellaars DJ. Knee strength deficits after hamstring tendon and patellar tendon anterior cruciate ligament reconstruction. Med Sci Sports Exerc. 2000;32(8):1472–9.
58. Huber R, Viecelli C, Bizzini M, et al. Knee extensor and flexor strength before and after anterior cruciate ligament reconstruction in a large sample of patients: influence of graft type. Phys Sportsmed. 2019;47(1):85–90.
59. Baumgart C, Welling W, Hoppe MW, Freiwald J, Gokeler A. Angle-specific analysis of isokinetic quadriceps and hamstring torques and ratios in patients after ACL-reconstruction. BMC Sports Sci Med Rehabil. 2018;10:23. https://doi.org/10.1186/s13102-018-0112-6 ((**eCollection 2018**)).
60. Angelozzi M, Madama M, Corsica C, et al. Rate of force development as an adjunctive outcome measure for return-to-sport decisions after anterior cruciate ligament reconstruction. J Orthop Sports Phys Ther. 2012;42(9):772–80.
61. Kadija M, Knezević OM, Milovanović D, et al. The effect of anterior cruciate ligament reconstruction on hamstring and quadriceps muscle function outcome ratios in male athletes. Srp Arh Celok Lek. 2016;144(3–4):151–7.
62. Hewett TE, Myer GD, Ford KR, et al. Biomechanical measures of neuromuscular control and valgus loading of the knee predict anterior cruciate ligament injury risk in female athletes: a prospective study. Am J Sports Med. 2005;33(4):492–501.
63. Krosshaug T, Steffen K, Kristianslund E, et al. The vertical drop jump is a poor screening test for ACL injuries in female elite soccer and handball players: a prospective cohort study of 710 athletes. Am J Sports Med. 2016;44(4):874–83.
64. Konrath JM, Vertullo CJ, Kennedy BA, et al. Morphologic characteristics and strength of the hamstring muscles remain altered at 2 years after use of a hamstring tendon graft in anterior cruciate ligament reconstruction. Am J Sports Med. 2016;44:2589–98.
65. Segawa H, Omori G, Koga Y, et al. Rotational muscle strength of the limb after anterior cruciate ligament reconstruction using semitendinosus and gracilis tendon. Arthroscopy. 2002;18(2):177–82.
66. Scanlan SF, Chaudhari AMW, Dyrby CO. Differences in tibial rotation during walking in ACL reconstructed and healthy contralateral knees. J Biomech. 2010;43(9):1817–22.

67. Noorkoiv M, Nosaka K, Blazevich AJ. Neuromuscular adaptations associated with knee joint angle-specific force change. Med Sci Sports Exerc. 2014;46(8):1525–37.
68. Brockett CL, Morgan DL, Proske U. Predicting hamstring strain injury in elite athletes. Med Sci Sports Exerc. 2004;36(3):379–87.
69. Morgan DL. New insights into the behavior of muscle during active lengthening. Biophys J. 1990;57(2):209–21.
70. Myer GD, Ford KR, Barber Foss KD, et al. The relationship of hamstrings and quadriceps strength to anterior cruciate ligament injury in female athletes. Clin J Sports Med. 2009;9(1):3–8.
71. Della Villa F, Andriolo L, Ricci M, et al. Compliance in postoperative rehabilitation is a key factor for return to sport after revision anterior cruciate ligament reconstruction. Knee Surg Sports Traumatol Arthrosc. 2020;28(2):463–9.
72. Tol JL, Hamilton B, Eirale C, et al. At return to play following hamstring injury the majority of professional football players have residual isokinetic deficits. Br J Sports Med. 2014;48:1364–9.
73. Wangensteen A, Tol JL, Witvrouw E, et al. Hamstring reinjuries occur at the same location and early after return to sport: a descriptive study of MRI-confirmed reinjuries. Am J Sports Med. 2016;44(8):2112–21.
74. Fyfe JJ, Opar DA, Williams MD, Shield AJ. The role of neuromuscular inhibition in hamstring strain injury recurrence. J Electromyogr Kinesiol. 2013;23(3):523–30.
75. Opar DA, Williams MD, Timmins RG, et al. Knee flexor strength and bicep femoris electromyographical activity is lower in previously strained hamstrings. J Electromyogr Kinesiol. 2013;23(3):696–703.
76. Opar DA, Williams MD, Timmins RG, et al. Eccentric hamstring strength and hamstring injury risk in Australian footballers. Med Sci Sports Exerc. 2014;47(4):857–65.
77. Choi JY, Ha JK, Kim YW, et al. Relationships among tendon regeneration on MRI, flexor strength, and functional performance after anterior cruciate ligament reconstruction with hamstring autograft. Am J Sports Med. 2012;40(1):152–62.
78. Snow BJ, Wilcox JJ, Burks RT, Greis PE. Evaluation of muscle size and fatty infiltration with MRI nine to eleven years following hamstring harvest for ACL reconstruction. J Bone Jt Surg Am. 2012;94:1274–82.
79. Papandrea P, Vulpiani MC, Ferretti A, Conteduca F. Regeneration of the semitendinosus tendon harvested for anterior cruciate ligament reconstruction evaluation using ultrasonography. Am J Sports Med. 2000;28:556–61.
80. Vertullo CJ, Konrath JM, Kennedy B, et al. Hamstring morphology and strength remain altered 2 years following a hamstring graft in ACL reconstruction. Orthop J Sports Med. 2017;5(5 suppl 5):2325967117S00181.
81. Irie K, Tomatsu T. Atrophy of semitendinosus and gracilis and flexor mechanism function after hamstring tendon harvest for anterior cruciate ligament reconstruction. Orthopedics. 2002;25:491–5.
82. Williams GN, Snyder-Mackler L, Barrance PJ, Axe MJ, Buchanan TS. Muscle and tendon morphology after reconstruction of the anterior cruciate ligament with autologous semitendinosus-gracilis graft. J Bone Jt Surg Am. 2004;86:1936–46.
83. Glasgow P, Phillips N, Bleakley C. Optimal loading: key variables and mechanisms. Br J Sports Med. 2015;49:278–9.
84. Lorenz DS, Reiman MP, Walker JC. Periodization: current review and suggested implementation for athletic rehabilitation. Sports Health. 2010;2:509–18.
85. Lorenz D, Morrison S. Current concepts in periodization of strength and conditioning for the sports physiotherapist. Int J Sports Phys Ther. 2015;10:734–47.
86. American College of Sports Medicine. Position stand: progression models in resistance training for healthy adults. Med Sci Sports Exerc. 2002;34:364–80.
87. Guex K, Degache F, Gremion G, Millet GP. Effect of hip flexion angle on hamstring optimum length after a single set of concentric contractions. J Sports Sci. 2013;31(14):1545–52.
88. McDonagh MJ, Davies CT. Adaptive response of mammalian skeletal muscle to exercise with high loads. Eur J Appl Physiol Occup Physiol. 1984;52(2):139–55.
89. Fry A. The role of resistance exercise intensity on muscle fibre adaptations. Sports Med. 2004;34:663–79.
90. Folland JP, Williams AG. The adaptations to strength training: morphological and neurological contributions to increased strength. Sports Med. 2007;37(2):145–68.
91. Anderson T, Kearney JT. Effects of three resistance training programs on muscular strength and absolute and relative endurance. Res Q Exerc Sport. 1982;53:1–7.
92. Campos GE, Luecke TJ, Wendeln HK, et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. Eur J Appl Physiol. 2002;88(1–2):50–60.
93. Harber MP, Fry AC, Rubin MR, et al. Skeletal muscle and hormonal adaptations to circuit weight training in untrained men. Scand J Med Sci Sports. 2004;14:176–85.
94. Aagaard P, Simonsen E, Andersen JL, et al. Neural inhibition during maximal eccentric and concentric quadricep contraction: effects of resistance training. J Appl Physiol. 2000;89:2249–57.
95. Andersen LL, Andersen JL, Kebis MK, Aagaard P. Early and late rate of force development: differential adaptive responses to resistance training? Scand J Med Sci Sports. 2010;20(1):e162–9.
96. Mangine GT, Hoffman JR, Wang R, et al. Resistance training intensity and volume affect changes in rate of force development in resistance-trained men. Eur J Appl Physiol. 2016;116:2367–74.
97. Tillin NA, Folland JP. Maximal and explosive strength training elicit distinct neuromuscular adaptations, specific to the training stimulus. Eur J Appl Physiol. 2014;114(12):365–74.
98. Schoenfeld BJ, Contreras B, Krieger J, et al. Resistance training volume enhances muscle hypertrophy but not strength in trained men. Med Sci Sports Exerc. 2019;51(1):94–103.
99. Burd NA, West DW, Staples AW, et al. Low-load high volume resistance exercise stimulates muscle protein synthesis more than low volume resistance exercise in young men. PLoS ONE. 2010;5:e12033.
100. Goto K, Ishii N, Kizuka T, et al. The impact of metabolic stress on hormonal responses and muscular adaptations. Med Sci Sports Exerc. 2005;37:955–63.
101. Cuthbert M, Ripley N, McMahon JJ, et al. The effect of Nordic hamstring exercise intervention volume on eccentric strength and muscle architecture adaptations: a systematic review and meta-analyses. Sports Med. 2020;50(1):83–99.
102. Buckthorpe M, La Rosa G, Villa FD. Restoring knee extensor strength after anterior cruciate ligament reconstruction: a clinical commentary. Int J Sports Phys Ther. 2019;14(1):159–72.
103. Reiman MP, Lorenz DS. Integration of strength and conditioning principles into a rehabilitation program. Int J Sports Phys Ther. 2011;6:241–53.
104. Carofino B, Fulkerson J. Medial hamstring tendon regeneration following harvest for anterior cruciate ligament reconstruction: fact, myth and clinical application. Arthroscopy. 2005;21:1257–65.
105. Escamilla RF, Macleod TD, Wilk KE, et al. Anterior cruciate ligament strain and tensile forces for weight-bearing and non-weight-bearing exercises: a guide to exercise selection. J Orthop Sports Phys Ther. 2012;42(3):208–20.

106. Ristanis S, Tsepis E, Giotis D, et al. Electromechanical delay of the knee lexor muscles is impaired after harvesting hamstring tendons for anterior cruciate ligament reconstruction. Am J Sports Med. 2009;37(11):2179–86.
107. Giles L, Webster KE, McClelland J, et al. Quadriceps strengthening with and without blood flow restriction in the treatment of patellofemoral pain: a double-blind randomised trial. Br J Sports Med. 2017;51:1688–94.
108. Whiteley R. Blood flow restriction training in rehabilitation: a useful adjunct or Lucy's latest trick? J Orthop Sports Phys Ther. 2019;49(5):294–8.
109. Della Villa S, Boldrini L, Ricci M, et al. Clinical outcomes and return-to-sports participation of 50 soccer players after Anterior Cruciate Ligament reconstruction through a sport-specific rehabilitation protocol. Sports Health. 2012;4(1):17–24.
110. Grindem H, Snyder-Mackler L, Moksnes H, et al. Simple decision rules can reduce reinjury risk by 84% after ACL reconstruction: the Delaware-Oslo ACL cohort study. Br J Sports Med. 2016;50:804–8.
111. Mjolsnes R, Arnason A, Osthagen T, et al. A 10-week randomized trial comparing eccentric vs. concentric hamstring strength training in well-trained soccer players. Scand J Med Sci Sports. 2004;14(5):311–7.
112. Seger JY, Arvidsson B, Thorstensson A. Specific effects of eccentric and concentric training on muscle strength and morphology in humans. Eur J Appl Physiol Occup Physiol. 1998;79(1):49–57.
113. Tomberlin JP, Basford JR, Schwen EE, et al. Comparative study of isokinetic eccentric and concentric quadriceps training. J Orthop Sports Phys Ther. 1991;14(1):31–6.
114. Boden BP, Dean GS, Feagin JA, et al. Mechanisms of anterior cruciate ligament injury. Orthopedics. 2000;23:573–8.
115. Chumanov ES, Schache AG, Heiderscheit BC, Thelen DG. Hamstrings are most susceptible to injury during the late swing phase of sprinting. Br J Sports Med. 2012;46(2):90.
116. Bourne M, Opar DA, Williams M, et al. Eccentric knee-flexor strength and hamstring injury risk in rugby union: a prospective study. Am J Sports Med. 2015;43(11):2663–70.
117. Fousekis K, Tsepis E, Poulmedis P, et al. Intrinsic risk factors of non-contact quadriceps and hamstring strains in soccer: a prospective study of 100 professional players. Br J Sports Med. 2011;45(9):709–14.
118. Timmins R, Bourne M, Shield A, et al. Strength and architectural risk factors for hamstring strain injury in elite Australian soccer: a prospective cohort study. J Sci Med Sport. 2015;19(Supplement):e20. https://doi.org/10.1016/j.jsams.2015.12.425.
119. van Dyk N, Bahr R, Whiteley R, et al. Hamstring and quadriceps isokinetic strength deficits are weak risk factors for hamstring strain injuries: a 4-year cohort study. Am J Sports Med. 2016;44(7):1789–95.
120. Bennell K, Wajswelner H, Lew P, et al. Isokinetic strength testing does not predict hamstring injury in Australian Rules footballers. Br J Sports Med. 1998;32(4):309–14.
121. Suchomel TJ, Wagle JP, Douglas J, et al. Implementing eccentric resistance training—part 1: a brief review of existing methods. J Funct Morphol Kinesiol. 2019;4(2):38.
122. Askling C, Karlsson J, Thorstensson A. Hamstring injury occurrence in elite soccer players after preseason strength training with eccentric overload. Scand J Med Sci Sports. 2003;13(4):244–50.
123. Iga J, Fruer CS, Deighan M, et al. 'Nordic' hamstrings exercise: engagement characteristics and training responses. Int J Sports Med. 2012;33(12):1000–4.
124. Brughelli M, Mendiguchia J, Nosaka K, et al. Effects of eccentric exercise on optimum length of the knee flexors and extensors during the preseason in professional soccer players. Phys Ther Sport. 2010;11(2):50–5.
125. Clark R, Bryant A, Culgan J, et al. The effects of eccentric hamstring strength training on dynamic jumping performance and isokinetic strength parameters: a pilot study on the implications for the prevention of hamstring injuries. Phys Ther Sport. 2005;6:67–73.
126. Kilgallon M, Donnelly AE, Shafat A. Progressive resistance training temporarily alters hamstring torque-angle relationship. Scand J Med Sci Sports. 2007;17(1):18–24.
127. Potier TG, Alexander CM, Seynnes OR. Effects of eccentric strength training on biceps femoris muscle architecture and knee joint range of movement. Eur J Appl Physiol. 2009;105(6):939–44.
128. de Breno AR, Alvares J, Marques VB, Vaz MA, et al. Four weeks of Nordic hamstring exercise reduce muscle injury risk factors in young adults. J Strength Cond Res. 2018;32(5):1254–62.
129. Presland J, Timmins R, Bourne M, et al. The effect of high or low volume Nordic hamstring exercise training on eccentric strength and biceps femoris long head architectural adaptations. Scand J Med Sci Sports. 2018;28:1775–83.
130. Alonso-Fernandez D, Docampo-Blanco P, Martinez-Fernandez J. Changes in muscle architecture of biceps femoris induced by eccentric strength training with Nordic hamstring exercise. Scand J Med Sci Sports. 2018;28(1):88–94.
131. Bourne MN, Timmins RG, Williams MD, et al. Impact of the Nordic hamstring and hip extension exercises on hamstring architecture and morphology: implications for injury prevention. Br J Sports Med. 2017;51(5):469–77.
132. Lovell R, Knox M, Weston M, et al. Hamstring injury prevention in soccer: before or after training? Scand J Med Sci Sports. 2018;28(2):658–66.
133. Pain MT, Forrester SE. Predicting maximum eccentric strength from surface EMG measurements. J Biomech. 2009;42:1598–603.
134. Blazevich AJ, Cannavan D, Coleman DR, Horne S. Influence of concentric and eccentric resistance training on architectural adaptation in human quadriceps muscles. J Appl Physiol. 2007;103(5):1565–75.
135. Crosier JL, Ganteaume S, Binet J, et al. Strength imbalance and prevention of hamstring injury in professional soccer players: a prospective study. Am J Sports Med. 2008;36:1469–75.
136. Arnason A, Andersen TE, Holme I, et al. Prevention of hamstring strains in elite soccer: an intervention study. Scand J Med Sci Sports. 2008;18:40–8.
137. Askling CM, Tengvar M, Tarassova O, Thorstensson A. Acute hamstring injuries in Swedish elite sprinters and jumpers: a prospective randomised controlled clinical trial comparing two rehabilitation protocols. Br J Sports Med. 2014;48:532–9.
138. Petersen J, Thorborg K, Nielsen MB, et al. Preventive effect of eccentric training on acute hamstring injuries in men's soccer: a cluster-randomized controlled trial. Am J Sports Med. 2011;39:2296–303.
139. van der Horst N, Smits DW, Petersen J, et al. The preventive effect of the Nordic hamstring exercise on hamstring injuries in amateur soccer players: a randomized controlled trial. Am J Sports Med. 2015;43(6):1316–23.
140. Novacheck TF. The biomechanics of running. Gait Posture. 1998;7:77–95.
141. Higashihara A, Nagano Y, Ono T, et al. Differences in hamstring activation characteristics between the acceleration and maximum-speed phases of sprinting. J Sports Sci. 2018;36:1313–28.
142. Hawkins D, Hull ML. A method for determining lower extremity muscle-tendon lengths during flexion/extension movements. J Biomech. 1990;23(5):487–94.
143. Visser JJ, Hoogkamer JE, Bobbert MF, et al. Length and moment arm of human leg muscles as a function of knee and hip-joint angles. Eur J Appl Physiol Occup Physiol. 1990;61:453–60.

144. Sugiura Y, Saito T, Sakuraba K, et al. Strength deficits identified with concentric action of the hip extensors and eccentric action of the hamstrings predispose to hamstring injury in elite sprinters. J Orthop Sport Phys Ther. 2008;38:457–64.
145. Powers CM. The influence of abnormal hip mechanics on knee injury: a biomechanical perspective. J Orthop Sports Phys Ther. 2010;40(2):42–51.
146. Wakahara T, Miyamoto N, Sugisaki N, et al. Association between regional differences in muscle activation in one session of resistance exercise and in muscle hypertrophy after resistance training. Eur J Appl Physiol. 2012;112(4):1569–76.
147. Wakahara T, Fukutani A, Kawakami Y, et al. Nonuniform muscle hypertrophy: its relation to muscle activation in training session. Med Sci Sports Exerc. 2013;45(11):2158–65.
148. Bourne MN, Williams MD, Opar DA, et al. Impact of exercise selection on hamstring muscle activation. Br J Sports Med. 2017;51:1021–8.
149. Mendiguchia J, Arcos AL, Garrues MA, et al. The use of MRI to evaluate posterior thigh muscle activity and damage during Nordic Hamstring exercise. J Strength Cond Res. 2013;27(12):3426–35.
150. Mendiguchia J, Garrues MA, Cronin JB, et al. Nonuniform changes in MRI measurements of the thigh muscles after two hamstring strengthening exercises. J Strength Cond Res. 2013;27(3):574–81.
151. Ono T, Okuwaki T, Fukubayashi T. Differences in activation patterns of knee flexor muscles during concentric and eccentric exercises. Res Sports Med. 2010;18(3):188–98.
152. Ono T, Higashihara A, Fukubayashi T. Hamstring functions during hip-extension exercise assessed with electromyography and magnetic resonance imaging. Res Sports Med. 2011;19(1):42–52.
153. Zebis MK, Skotte J, Andersen CH, et al. Kettlebell swing targets semitendinosus and supine leg curl targets biceps femoris: an EMG study with rehabilitation implications. Br J Sports Med. 2013;47(18):1192–8.
154. Fernandez-Gonzalo R, Tesch PA, Linnehan RM, et al. Individual muscle use in hamstring exercises by soccer players assessed using functional MRI. Int J Sports Med. 2016;37(7):559–64.
155. Messer DJ, Bourne MN, Williams MD, et al. Hamstring muscle use in women during hip extension and the Nordic hamstring exercise: a functional magnetic resonance imaging study. J Orthop Sports Phys Ther. 2018;48(8):607–12.
156. Lynn SK, Costigan PA. Changes in the medial-lateral hamstring activation ratio with foot rotation during lower limb exercise. J Electromyogr Kinesiol. 2009;19(3):e197-205.
157. Buckthorpe M, Gimpel M, Wright S, et al. Hamstring muscle injuries in elite football: translating research into practice. Br J Sports Med. 2018;52(10):628–9.
158. Sale DG. Neural adaptations to resistance training. Med Sci Sports Exerc. 1988;20(Supplement 5):S135–45.
159. Buckthorpe MW, Erskine R, Fletcher G, Folland F. Neural adaptations explain the task specificity of strength changes after resistance training. Scand J Med Sci Sports. 2015;25:640–9.
160. Cacchio A, Don R, Ranavolo A, et al. Effects of 8-week strength training with two models of chest press machines on muscular activity pattern and strength. J Electromyogr Kinesiol. 2008;18:618–27.
161. Davis IS, Powers CM. Patellafemoral pain syndrome: proximal, distal and local factors, an international retreat. April 30-May 2, 2009, Fells Point, Baltimore, MD. J Orthop Sports Phys Ther. 2010;40(3):A1–16.
162. Ireland ML, Willson JD, Ballantyne BT, et al. Hip strength in females with and without patellafemoral pain. J Orthop Sports Phys Ther. 2003;33(11):671–6.
163. Khayambashi K, Ghoddosi N, Straub RK, Powers CM. Hip muscle strength predicts non-contact anterior cruciate ligament injury in male and female athletes: a prospective study. Am J Sports Med. 2016;44(2):355–61.
164. Leetun DT, Ireland ML, Willson JD, et al. Core stability measures as risk factors for lower extremity injury in athletes. Med Sci Sports Exerc. 2004;36(6):926–34.
165. Zakulak BT, Hewett TE, Reeves NP, et al. Deficits in neuromuscular control of the trunk predict knee injury risk: a prospective biomechanical–epidemiological study. Am J Sports Med. 2007;35(7):1123–30.
166. Zakulak BT, Hewett TE, Reeves NP, et al. The effects of core proprioception on knee injury: a prospective biomechanical–epidemiological study. Am J Sports Med. 2007;35(3):368–73.
167. Kuszewski M, Gnat R, Saulicz E. Stability training of the lumbo-pelvo-hip complex influence stiffness of the hamstrings: a preliminary study. Scand J Med Sci Sports. 2009;19:260–6.
168. Schuermans J, Danneels L, Tiggelen DV, et al. Proximal neuromuscular control protects against hamstring injuries in male soccer players: a prospective study with electromyography time-series analysis during maximal sprinting. Am J Sports Med. 2017;45:1315–25.
169. Mills M, Frank B, Goto S, et al. Effects of restricted hip flexor muscle length on hip extensor muscle activity and lower extremity biomechanics in college-aged female soccer players. Int J Sports Phys Ther. 2015;10:946–54.
170. Mendiguchia J, Alentorn-Geli E, Brughelli M. Hamstring strain injuries: are we heading in the right direction? Br J Sports Med. 2012;46(2):81–5.
171. Decker MJ, Torry MR, Noonan TJ, Riviere A, Sterett WI. Landing adaptations after ACL reconstruction. Med Sci Sports Exerc. 2002;34(9):1408–13.
172. de Fontenay BP, Argaud S, Blache Y, Monteil K. Motion alterations after anterior cruciate ligament reconstruction: comparison of the injured and uninjured lower limbs during a single-legged jump. J Athl Train. 2014;49(3):311–6.
173. Goerger BM, Marshall SW, Beutler AI. Anterior cruciate ligament injury alters preinjury lower extremity biomechanics in the injured and uninjured leg: the JUMP-ACL study. Br J Sports Med. 2015;49:188–95.
174. Lee SP, Chow JW, Tillman MD. Persons with reconstructed ACL exhibit altered knee mechanics during high speed maneuvers. J Sports Med. 2014;35(6):528–33.
175. Paterno MV, Ford KR, Myer GD, Heyl R, Hewett TE. Limb asymmetries in landing and jumping 2 years following anterior cruciate ligament reconstruction. Clin J Sports Med. 2007;17(4):258–62.
176. Sterns KM, Pollard CD. Abnormal frontal plane knee mechanics during sidestep cutting in female soccer athletes after anterior cruciate ligament reconstruction and return to sport. Am J Sports Med. 2013;41(4):918–23.
177. Paterno MV, Schmitt LC, Ford KR, et al. Biomechanical measures during landing and postural stability predict second anterior cruciate ligament after anterior cruciate ligament reconstruction and return to sport. Am J Sports Med. 2010;38(10):1968–78.
178. Paterno MV, Kiefer AW, Bonnette S, et al. Prospectively identified deficits in sagittal plane hip-ankle coordination in female athletes who sustain a second anterior cruciate ligament injury after anterior cruciate ligament reconstruction and return to sport. Clin Biomech. 2015;30(10):1094–104.
179. Hruska R. Pelvic stability influences lower extremity kinematics. Biomech. 1998;6:23–9.
180. Loudon JK, Jenkins W, Loudon KL. The relationship between static posture and ACL injury in female athletes. J Orthop Sports Phys Ther. 1996;24(2):91–7.
181. Dill KE, Begalle RL, Frank BS, et al. Altered knee and ankle kinematics during squatting in those with limited

weight-bearing–lunge ankle-dorsiflexion range of motion. J Athl Train. 2014;49(6):723–32.
182. Palmieri-Smith RM, Lepley LK. Quadriceps strength asymmetry following ACL reconstruction alters knee joint biomechanics and functional performance at time of return to activity. Am J Sports Med. 2015;43(7):1662–9.
183. Sahrmann S. Diagnosis and treatment of movement impairment syndromes. Oxford: Elsevier Health Sciences; 2013.
184. Aman JE, Elangovan N, Yeh IL, Konczak J. The effectiveness of proprioceptive training for improving motor function: a systematic review. Front Hum Neurosci. 2014;8:1075.
185. Anderson K, Behm DG. The impact of instability resistance training on balance and stability. Sports Med. 2005;35(1):43–53.
186. Zebis MK, Andersen LL, Bencke J, et al. Identification of athletes at future risk of anterior cruciate ligament ruptures by neuromuscular screening. Am J Sports Med. 2009;37(10):1967–73.
187. Krosshaug T, Nakamae A, Boden BP, et al. Mechanisms of anterior cruciate ligament injury in basketball: video analysis of 39 cases. Am J Sports Med. 2007;35:359–67.
188. Dyhre-Poulsen P, Krogsgaard MR. Muscular reflexes elicited by electrical stimulation of the anterior cruciate ligament in humans. J Appl Physiol (1985). 2000;89(6):2191–5.
189. Zebis MK, Andersen LL, Bencke J, et al. The effects of neuromuscular training on knee joint motor control during sidecutting in female elite soccer and handball players. Clin J Sport Med. 2008;18(4):329–37.
190. Della Villa F, Buckthorpe M, Grassi A, et al. Systematic video analysis of ACL injuries in professional male football (soccer): injury mechanisms, situational patterns and biomechanics study on 134 consecutive cases. Br J Sports Med. 2020. https://doi.org/10.1136/bjsports-2019-101247 ((**published online first**)).
191. Koga H, Nakamae A, Shima Y, et al. Mechanisms for noncontact anterior cruciate ligament injuries: knee joint kinematics in 10 injury situations from female team handball and basketball. Am J Sports Med. 2010;38:2218–25.
192. Waldén M, Krosshaug T, Bjørneboe J, et al. Three distinct mechanisms predominate in non-contact anterior cruciate ligament injuries in male professional football players: a systematic video analysis of 39 cases. Br J Sports Med. 2015;49:1452–60.
193. Webster KE, Hewett TE. Meta-analysis of meta-analyses of anterior cruciate ligament injury reduction training programs. J Orthop Res. 2018;36(10):2696–708.
194. Tillin NA, Pain MTG, Folland JP. Short-term unilateral resistance training affects the agonist-antagonist but not the force-agonist activation relationship. Muscle Nerve. 2011;43:375–84.
195. Thorstensson A, Karlsson J, Viitasalo HT, et al. Effect of strength training on EMG of human skeletal muscle. Acta Physiol Scand. 1976;98:232–6.
196. Beneke R, Taylor MJ. What gives Bolt the edge-A.V. Hill knew it already! J Biomech. 2010;43(11):2241–3.
197. Kivi DM, Maraj BK, Gervais P. A kinematic analysis of high-speed treadmill sprinting over a range of velocities. Med Sci Sports Exerc. 2002;34(4):662–6.
198. Folland JP, Buckthorpe MW, Hannah R. Human capacity for explosive force production: neural and contractile determinants. Scand J Med Sci Sports. 2014;24(6):894–906.
199. Buckthorpe M, Roi GS. The time has come to incorporate a greater focus on rate of force development training in the sports injury rehabilitation process. Muscles Ligaments Tendons J. 2018;7(3):435–41.
200. Buckthorpe M, Wright S, Bruce-Low S, et al. Recommendations for hamstring injury prevention in elite football: translating research into practice. Br J Sports Med. 2019;53:449–56.
201. Bobbert MF, Van Soest AJ. Effects of muscle strengthening on vertical jump height: a simulation study. Med Sci Sports Ex. 1994;26(8):1012–20.
202. Herman DC, Weinhold PS, Guskiewicz KM, Garrett WE, Yu B, Padua DA. The effects of strength training on the lower extremity biomechanics of female recreational athletes during a stop-jump task. Am J Sports Med. 2008;36(4):733–40.
203. Nagano A, Gerritsen KGM. Effects of neuromuscular strength training on vertical jumping—a computer simulation study. J Appl Biomech. 2001;17(2):113–28.
204. Buckthorpe M, Stride M, Della VF. Assessing and treating gluteus maximus weakness—a clinical commentary. Int J Sports Phys Ther. 2019;14(4):655–69.
205. Paterno MV, Rauh MJ, Schmitt LC, Ford KR, Hewett TE. Incidence of second ACL injuries 2 years after primary ACL reconstruction and return to sport. Am J Sports Med. 2014;42(7):1567–73.
206. Andrade R, Pereira R, van Cingel R, et al. How should clinicians rehabilitate patients after ACL reconstruction? A systematic review of clinical practice guidelines (CPGs) with a focus on quality appraisal (AGREE II). Br J Sports Med. 2020;54(9):512–9.
207. van Melick N, van Cingel RE, Brooijmans F, et al. Evidence-based clinical practice update: practice guidelines for anterior cruciate ligament rehabilitation based on a systematic review and multidisciplinary consensus. Br J Sports Med. 2016;50(24):1506–15.
208. Herrington L, Myer G, Horsley I. Task based rehabilitation protocol for elite athletes following anterior cruciate ligament reconstruction: a clinical commentary. Phys Ther Sport. 2013;14(4):188–98.
209. Brukner P, Nealon A, Morgan C, et al. Recurrent hamstring muscle injury: applying the limited evidence in the professional football setting with a seven-point programme. Br J Sports Med. 2014;48:929–38.
210. Oakley AJ, Jennings J, Bishop CJ. Holistic hamstring health: not just the Nordic hamstring exercise. Br J Sports Med. 2018;52(13):816–7.
211. Freeman BW, Young WB, Talpey SW, et al. The effects of sprint training and the Nordic hamstring exercise on eccentric hamstring strength and sprint performance in adolescent athletes. J Sports Med Phys Fit. 2019;59(7):1119–25.
212. van den Tillaar R, Solheim JAB, Bencke J. Comparison of hamstring muscle activation during high-speed running and various hamstring strengthening exercises. Int J Sports Phys Ther. 2017;12:718–27.
213. Dorn TW, Schache AG, Pandy MG. Muscular strategy shift in human running: dependence of running speed on hip and ankle muscle performance. J Exp Biol. 2012;215(Pt 11):1944–56.
214. Chumanov ES, Heiderscheit BC, Thelen DG. The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of running. J Biomech. 2007;40:3555–62.
215. Woods C, Hawkins RD, Maltby S, et al. The Football Association Medical Research Programme: an audit of injuries in professional football- analysis of hamstring injuries. Br J Sports Med. 2004;38:36–41.
216. Barber-Westin SD, Noyes FR. Factors used to determine return to unrestricted sports activities after anterior cruciate ligament reconstruction. Arthroscopy. 2011;27(12):1697–705.
217. Timmins R, Bourne M, Shield A, et al. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. Br J Sports Med. 2015;50(24):1524–35.